UNIT-1 INTRODUCTION TO PC HARDWARE

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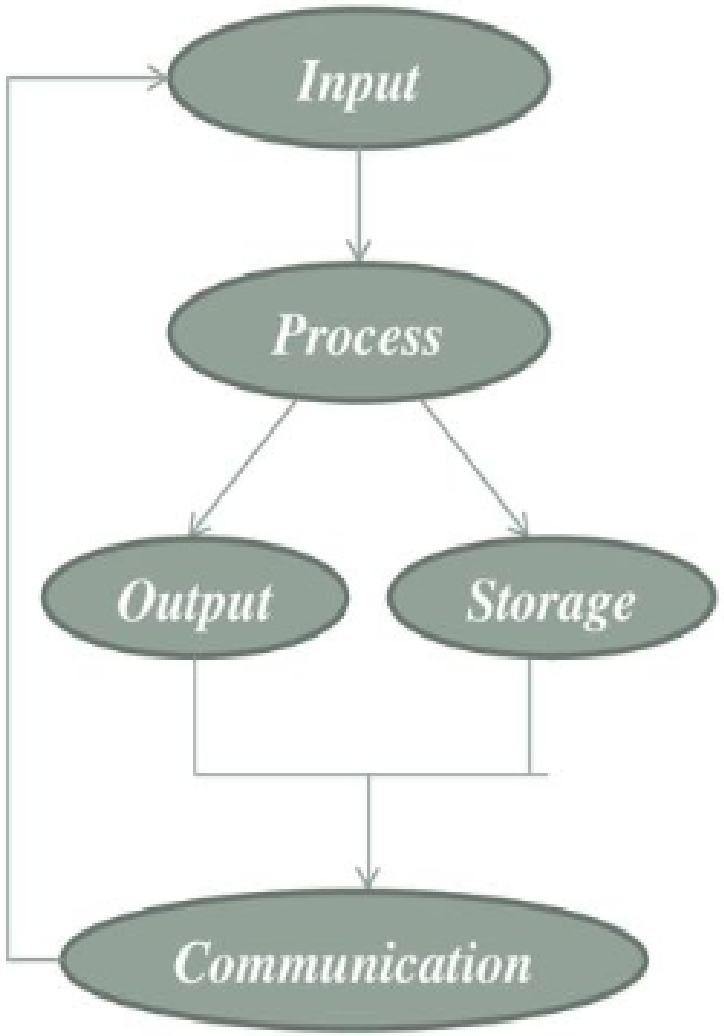
Topics to be Discussed

- Definition of Computer
- Computer Hardware
- Software
- Firmware
- History of Computer
- Classification of Computer

What is A Computer?

- An electronic machine/device that **manipulates information**, or **data**.
- Ability to **store**, **retrieve**, and **process data**.

- Input-Process-Output
  - Capable of solving problem or manipulating data by accepting data
  - Performing mathematical or logical operations on data.
  - Supplying the result of these operations as output.



Characteristics of Computer

- **Speed**
  - Quick performance & quick data input and information retrieval.
- **Storage Capacity**
  - Ability to store data.
- **Accuracy**
  - Computer is programmed, so whatever input we give it gives result accurately.
- **No IQ**
  - Computer does not work without instruction.
- **Versatility**
  - Perform completely different type of work at the same time.

What is Hardware?

Hardware represents the physical and tangible components of a computer i.e. the components that can be seen and touched.

Examples:
Input devices -- keyboard, mouse etc.
Output devices -- printer, monitor etc.
Secondary storage devices -- Hard disk, CD, DVD etc.
Internal components -- CPU, motherboard, RAM etc.

What is Firmware?

- It is **pre-written program**, which is stored in **ROM**. It is added at the time of manufacturing, which is used to run programs on the device.

- Its **combination** of hardware and software.

- ROMs, PROMs and EPROMs that have data or programs recorded on them are firmware.

What is Software?

Software is a set of programs, which is designed to perform a well-defined function. A program is a sequence of instructions written to solve a particular problem.

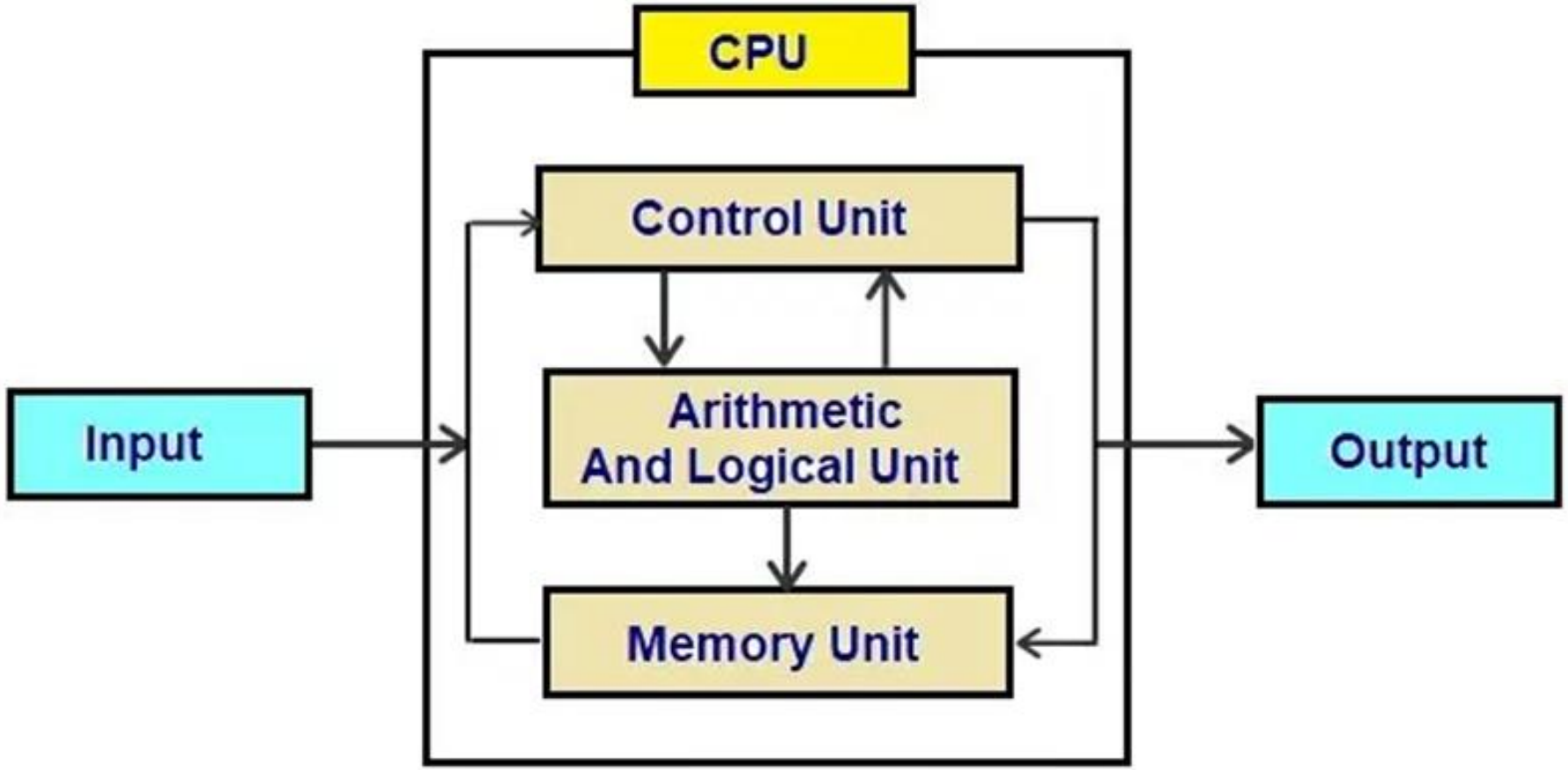
A software acts as an interface between the user and the hardware.

There are two types of software

- System Software
- Application Software

Some of those components, like Windows XP, Microsoft Word, Microsoft Excel, Windows Media Player, and all your other programs, are software.

PC Block Diagram:



Block Diagram of a Computer

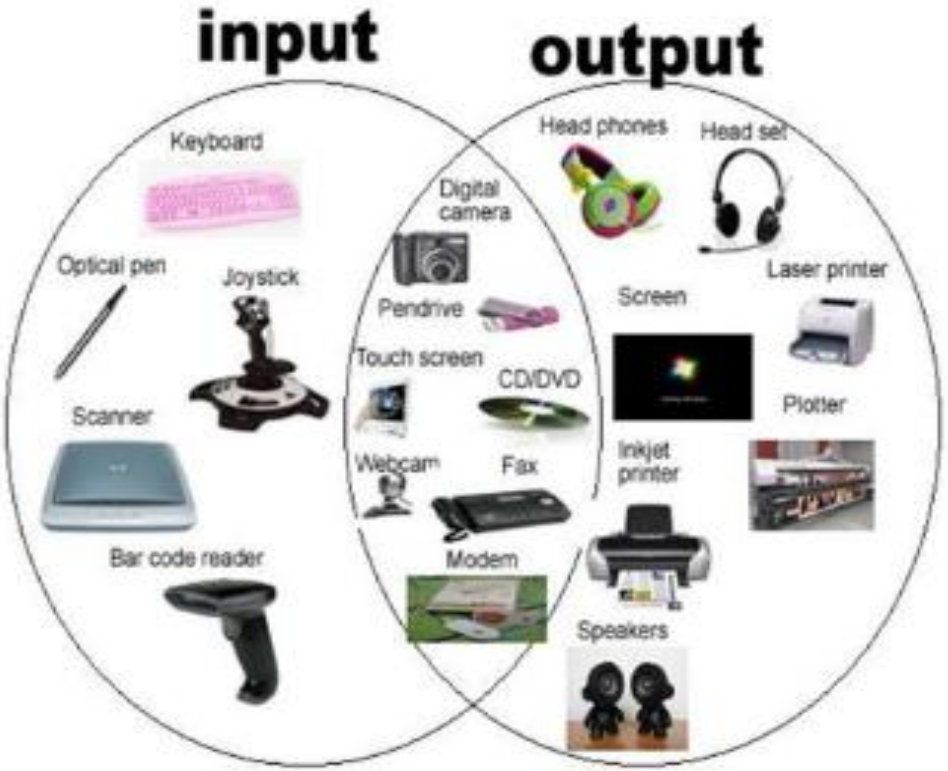
The Components of A Computer

Input Device:

- It is a hardware use to enter data and instructions.
- Converts the data into machine-readable form i.e binary form.

Output Device:

- It is a hardware that conveys information to the user.
- Converts the electronic data produced by the computer system.



The Components of A Computer

System Unit:

- Box-like case containing electronic components used to process data.
- Pocessor, Memory, Adaptar Cards, Ports, Power Supply

Monitor

A computer monitor is an electronic device that shows pictures. Monitors often look similar to televisions.

There are **three types** of computer displays:

1. CRT monitor (**Cathode Ray Tube**)
2. LCD monitor (**Liquid Crystal Display**)
3. Video Projectors

CPU

- A Central Processing Unit (CPU) is a component in digital Computer that interprets computer program instructions and process data.

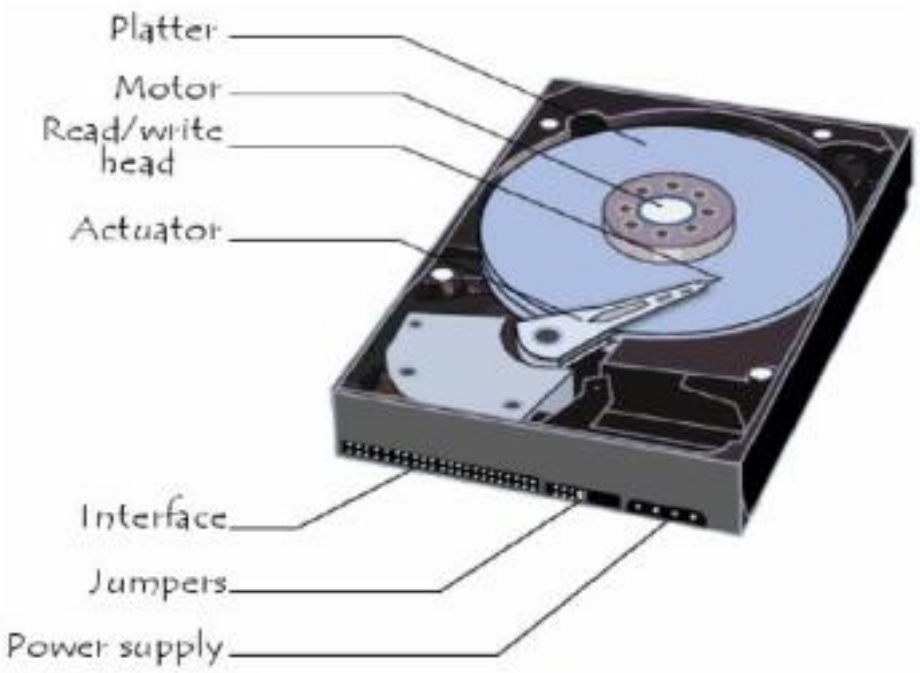
- 2 Major Manufactures are there:
  - Intel : http://www.intel.com
  - AMD : http://www.amd.com

Optical Disk Drive

- An optical drive is usually a CD drive or DVD drive.
- The drive uses a light called a laser. A laser is the most exact and powerful sort of light but the laser in the drive is very, very small.
- Types of Optical Drive
  - CD Drive
  - DVD Drive
  - Blue-ray Drive
  - HD-DVD Drive

Hard Disk Drive

- A hard disk drive (HDD), is something used by computers to store information.
- Hard disks use magnetic recording (similar to the way recording is done on magnetic tapes) to store information on rotating circular platters.
- The capacity of a hard drive is usually measured in gigabytes (GB).



Keyboard

- A computer keyboard is an important device that allows a person to enter symbols like letters and numbers into a computer.

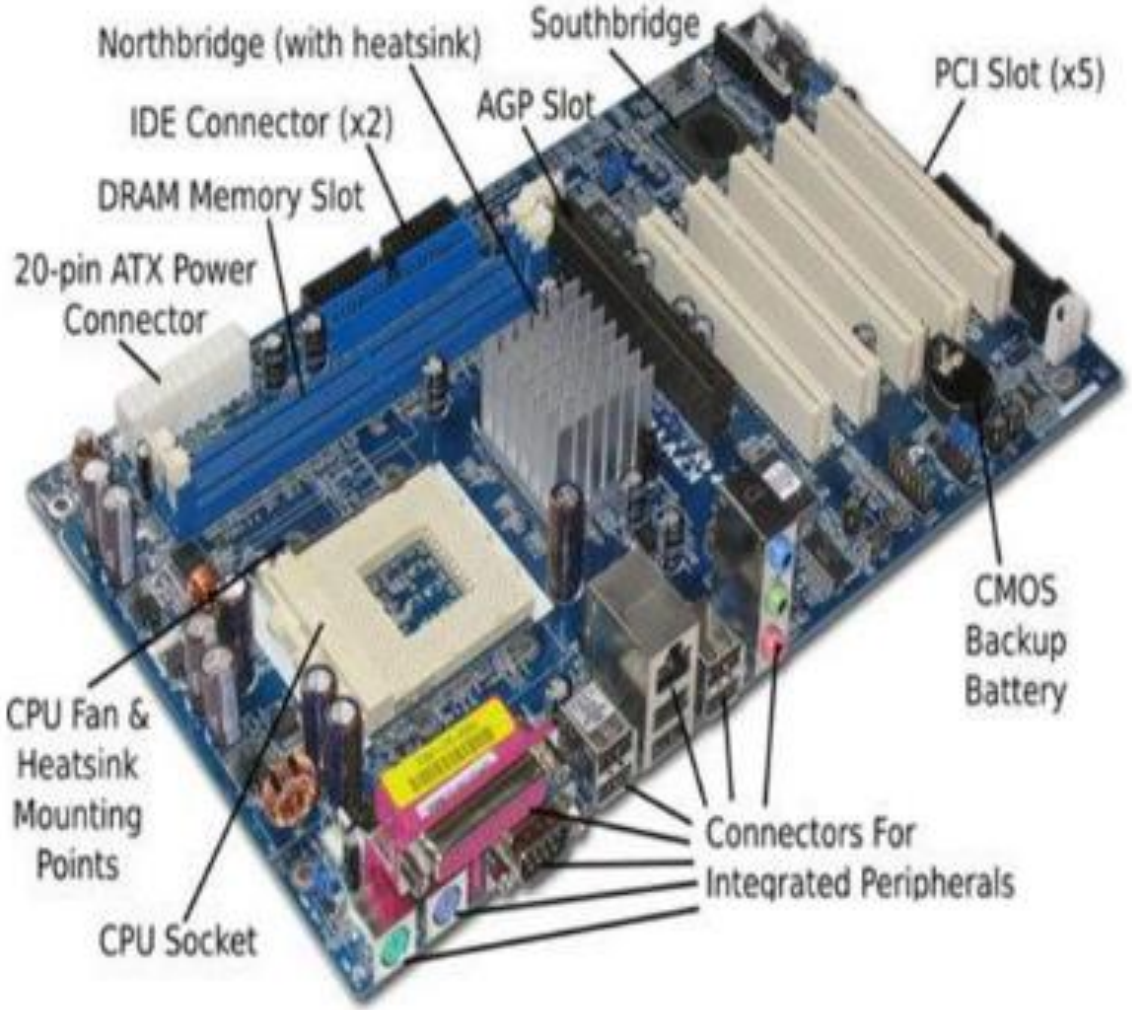
- It is the main input device for most computers.

Mouse

- A **computer mouse** is an input device that is most often used with a personal computer.
- Moving a mouse along a flat surface can move the on-screen cursor to different items on the screen.
- Items can be moved or selected by pressing the mouse buttons (called clicking).

Motherboard

- The motherboard is the main circuit board in a system, like a computer. It is the most 'central' part of a computer.
- All of the different parts of the computer are connected to the motherboard.
- Contains adaptar cards,processor chips and memory chips.
- In most computers, the motherboard is a big green board, but many come in different colors like black, red and yellow.



Processor

What is central processing unit (CPU)?

- It interprets and carries out basic instructions that operate a computer.

Control Unit:

Directs and Co-ordinates operations in computer.

Arithmetic Logical Unit:

ALU performs arithmetic, comparison and logical operations.

Memory

Memory is used to store data and instructions.

Cache Memory:

- Cache memory is a very high speed semiconductor memory which can speed up CPU. It acts as a buffer between the CPU and main memory. It is used to **hold** those parts of data and program which are **most frequently used** by CPU.

Primary Memory: RAM,ROM

- Stores all or part of the programs that is being executed

  - **RAM(Random Access Memory)**
    - It is the internal memory of the CPU for storing data, program and its result.
    - Its contents will be lost when the computer is turned off.

Cont..

ROM (Read Only Memory)

- The memory from which we can only read **but cannot write** on it. This type of memory is **non-volatile.**
- Stores important or frequently used programs.

Secondary Memory: Storage Devices

- Devices that store information.
- Stores important or frequently used programs.
- **Example :** Hard Disk Drive, CD and DVD drives, USB drives

Classification Of Computer

Types of computer

Mini computer

Super computer

Workstation

Micro computer

Mainframe computer

Micro Computer

- The microcomputers are also known as personal computers or simply PCs.
- They are also called Desktop computers. Microcomputers are the most common, smallest and very cheaper computers.
- They were introduced in 1970's as a result of the development of microprocessor.
- A single microprocessor may carry more than hundred thousand transistors on a single silicon chip.

Micro Computer

Size:

A personal computer can be placed easily on any table or desk.

Desktop PCs

Micro Computer

Uses

- They are being widely accepted and used in business, at home and in educational institutions.
- They are commonly known as personal computers and are usually single user machines but can be interconnected among themselves to form a Local Area Network.

Mainframe Computer

- A mainframe (also known as "big iron") is a high-performance computer used for large-scale computing purposes that require greater availability and security of data than a smaller-scale machine can offer.
- In the past, a typical mainframe might have occupied 2,000 - 10,000 square feet. Now the mainframe computer specially requires a very large clean room with air-conditioner.

Mainframe Computer

Operating System

- Special operating systems are used on main frames such as z/VM, z/VSE™, Linux for System z™, and z/TPF.

Mainframe Computer

Uses:

- The mainframe computers are specially used as servers on the World Wide Web. The mainframe computers are used in large organizations such as Banks, Airlines and Universities etc. where many people (users) need frequent access to the same data, which is usually organized into one or more huge databases.

Examples:

IBM S/390, Control Data CYBER 176 and Amdahl 580

Miniframe Computer

- Minicomputers were first introduced in early 1960s. A minicomputer is a class of multi-user computers that lies in the middle range of the mainframe computers and the microcomputers.
- They are smaller than mainframes in size and other facilities, such as speed, storage capacity and services.

Size:

A minicomputer and its peripheral equipment can usually fill a small room. A minicomputer typically about the size and shape of a wardrobe, mounted in a single tall rack.

Miniframe Computer

Working:

- Minicomputers are commonly used as servers like a small mainframe in a network environment and hundreds of users can be connected to the network with a minicomputer at a time.
- Now the terms "midrange computer" and "server" replaced the venerable minicomputer designation.

Miniframe Computer

Uses

The minicomputers are used in business, education and many other government departments. They are also widely used in industrial and small business application.

Miniframe Computer

Examples:

IBM's AS 400, Hewlett Packard HP 3000 series, Data General's MV/1500 etc.

Super Computer

- Super computer is a large computer or collection of computers that act as one large computer capable of processing enormous amounts of data. They are the fastest and most powerful type of computers.
- ***Size:***

A minicomputer and its peripheral equipment can usually fill a small room. A minicomputer typically about the size and shape of a wardrobe, mounted in a single tall rack.

Super Computer

Working:

- A super computer can execute one instruction in Pico-seconds (thousand-billionth of a second). Mostly they are based on the principle of parallel processing. They have extraordinary high speed due to the use of several processors connected in parallel.
- Columbia is a supercomputer built by Silicon Graphics for NASA. Its main purpose was to simulate the violent collision and merger of spiral galaxies.

Super Computer

- It is composed of 10,240 processors. It has 20 terabytes of Random-access memory, 440 terabytes of storage, and 10 petabytes of archive storage.

Super Computer

Uses

Super computers are normally used in large organizations, research laboratories, aerospace centers, large industrial units, atomic reactors or the places where very high processing speed is required.

Super Computer

Examples:

Cray 1, Cray 2 Cray XMP and CYBER 205

Workstation Computer

- A **workstation** refers to an individual computer, or group of computers, used by a single user to perform work.
- For example, a "workstation" may be an average-powered computer connected to a larger network. It can also refer to a powerful computer intended for serious academic or professional computation.
- **Example usage**
- In the sentence "At Pixar, the workstations are all Intel-based," the term "workstations" refers to powerful 3D modeling and animation computers. It would not refer to computers in Pixar's render farm. Although render farm computers are extremely powerful, they are not intended for a human to use directly or individually.
- **History**
- The Xerox Alto, introduced in 1974 and considered to be the first workstation.

❖ Assignment – Latest Specification of Laptop/computer

- OS -?
- Processor-?
- Memory(RAM)-?
- Storage- ?
- Monitor/Display-?
- Network Adaptor-?
- Camera-?
- Other?

Various types of ports and connectors found in PC/laptop

- A computer is a device that transforms data into meaningful information. It processes the input according to the set of instructions provided to it by the user and gives the desired output.
- As we know that we can connect multiple external devices with the computer system. Now, these devices are connected with the computer using Ports.
- The ports are the physical docking points present in the computer through which the external devices are connected using cables. Or in other words, a port is an interface between the motherboard and an external device of the computer.
- There are different types of ports available:

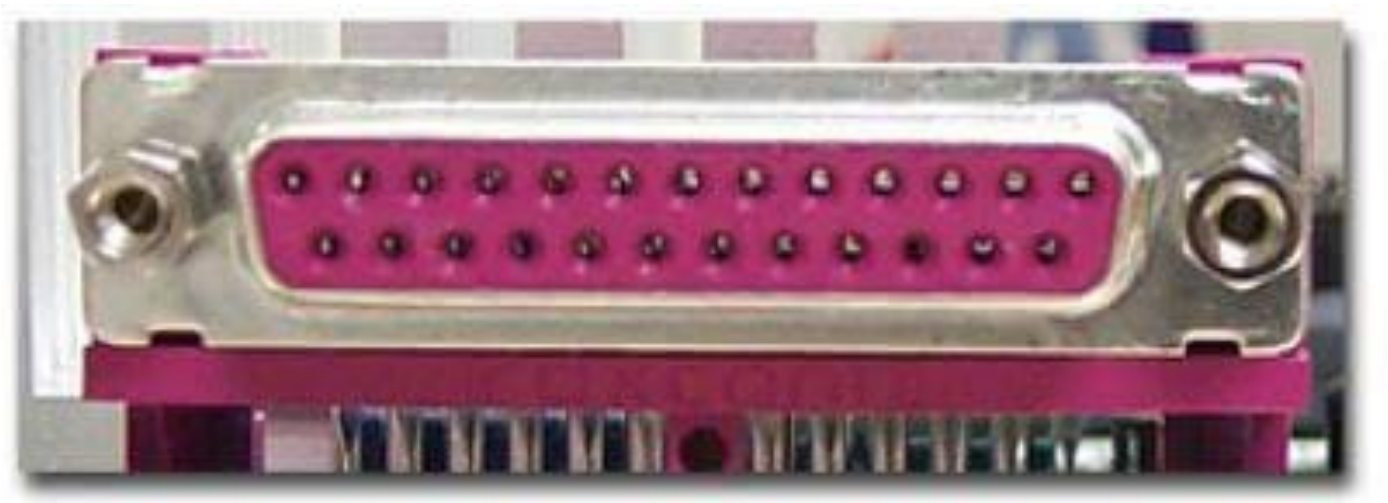
1. Serial port
2. Parallel port
3. USB port
4. PS/2 port
5. VGA port
6. Modem port
7. FireWire Port
8. Sockets
9. Infrared Port
10. Game Port
11. Digital Video Interface(DVI) Port
12. Ethernet Port
13. HDMI Port

1. Serial port(COM Port):

- A serial port is also called a communication port and they are used for connection of external devices like a modem, mouse, or keyboard (basically in older PCs).
- Serial cables are cheaper to make in comparison to parallel cables and they are easier to shield from interference. There are two versions of it, which are 9 pin model and 25 pin model. It transmits data at 115 KB/sec.

2. Parallel Port (LPT ports):

- Parallel ports are generally used for connecting scanners and printers. It can send several bits at the same time as it uses parallel communication.
- Its data transfer speed is much higher in comparison with the serial port. It is a 25 pin model. It is also known as Printer Port or Line Printer Port.



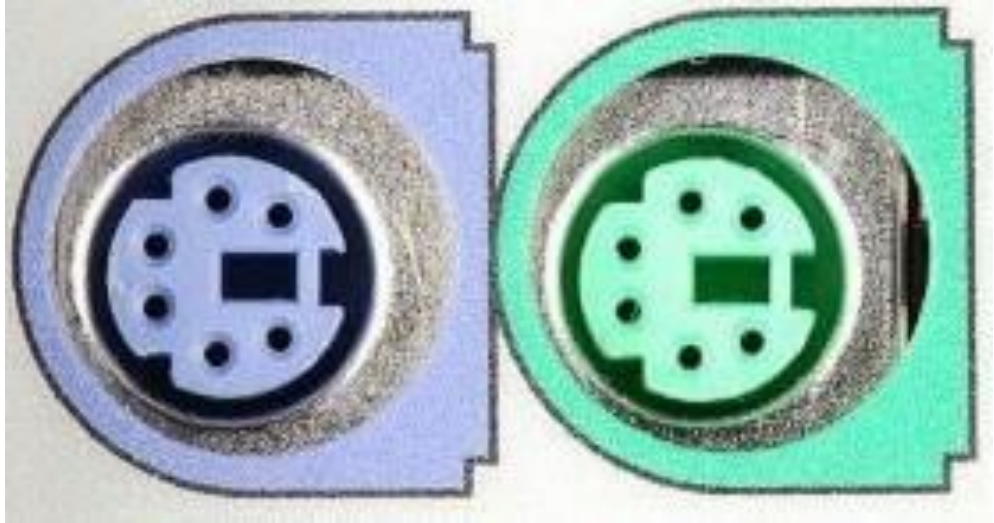
3. USB (Universal Serial Bus):

- In 1997 USB was first introduced. This can connect all kinds of external USB devices, like external hard disk, printer, scanner, mouse, keyboard, etc.
- There are minimum of two USB Ports provided in most of the computer systems. It is a kind of new type serial connection Port that is much faster than the old serial Ports and These USB Ports are much smarter and more versatile, as it allows the “daisy chaining” of up to 127 USB peripherals connected to one port.
- The data transfer rate in this is Data12 megabits per second. It also provides plug & plays communication.

USB type C

4. PS/2 Port:

- PS/2 ports are special ports used for connecting old computer keyboard and mouse.
- It was invented by IBM. In old computers, there are minimum of two PS/2 Ports, each for the keyboard and the mouse. It is a 6 pin mini Din connector.



5. VGA Port:

- VGA ports also known as Video Graphic Array connector are those which connect the monitor to a computer's video card.

- VGA port has 15 holes and it is similar to the serial port connector. But VGA Ports have holes in it and the serial port connector has pins in it.
- This port is commonly found in computers, projectors, and high definition TVs.

6. Sockets:

- Microphones and speakers are connected with the help of Sockets to the sound card of the computer.

7. FireWire Port:

- The IEEE 1394 interface, which is developed in the late 1980s and early 1990s by Apple as FireWire. It can transfer large amount of data at very high speed. It is used to connect camcorders and video equipment to the computer. It comes up with three variants which are 4-Pin FireWire 400 connector, 6-Pin FireWire 400 connector, and 9-Pin FireWire 800 connector.

8. Infrared Port:

- An Infrared(IR) port is used to sends and receives infrared signals from other devices. It is a kind of wireless type port with a limited range of 5-10ft.

9. Game Port:

- These ports are used previously to connect a joystick to a PC. But nowadays it is replaced by USB ports.

10. Modem Port:

- As the name suggests, a Modem port is used to connects a PC's modem to the telephone network.

11. Digital Video Interface(DVI) Port:

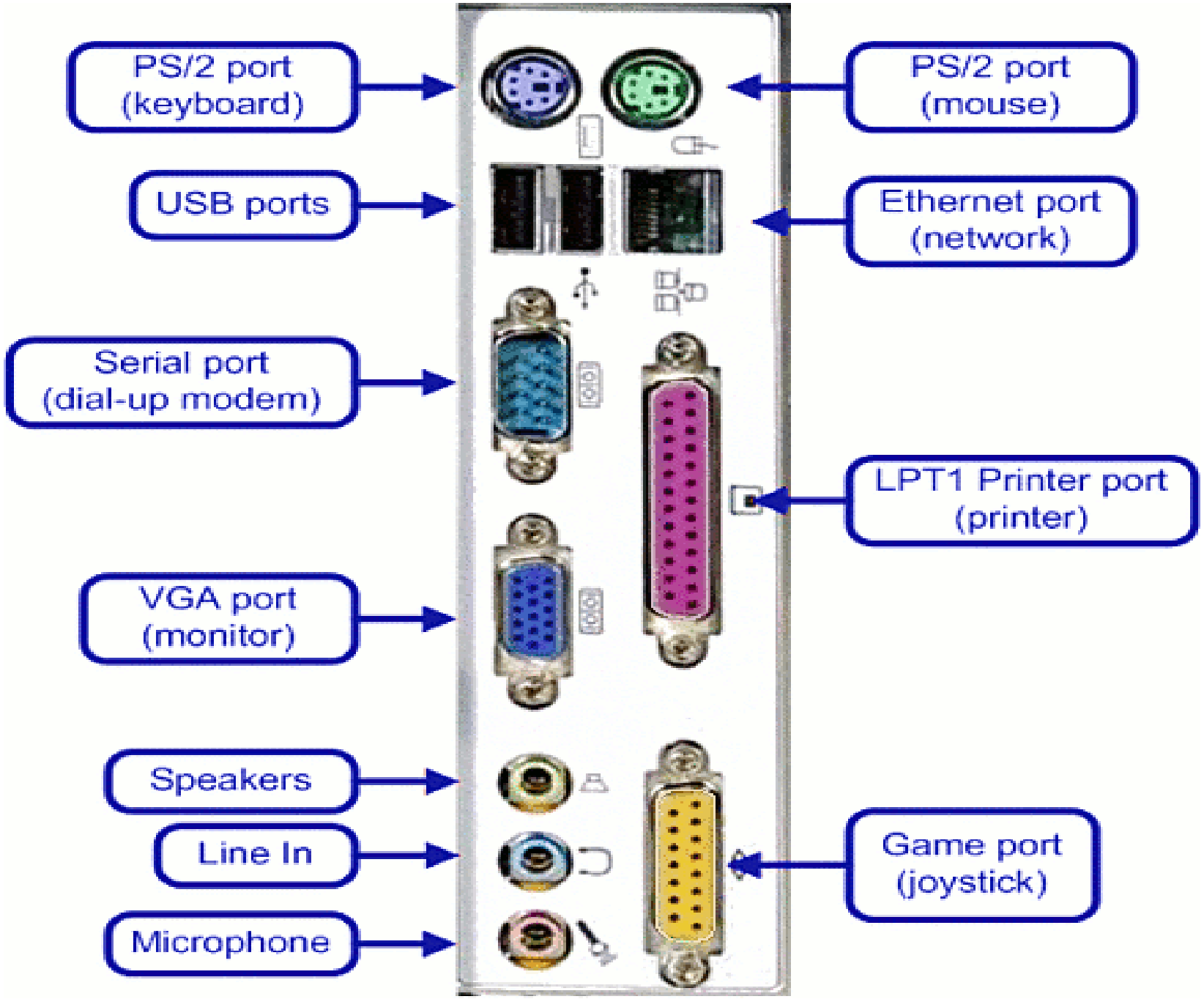
- DVI Port is used to connects LCD(flat panel) monitor to the computer's high-end video graphic cards and it is very popular among video card manufacturers.

12. Ethernet Port:

- Ethernet Port helps to connect to a network and high-speed Internet(provided by LAN or other sources).
- It connects the network cable to a computer and resides in a Ethernet card. It provides a data travel speed of 10 Mb to 1000 Mb(megabits) per second.

13. HDMI port:

- HDMI (High Definition Media Interface) is a digital interface developed to connect high definition devices such as digital cameras, gaming consoles, etc., to computers and TVs with HDMI ports.
- Besides this, it can carry uncompressed video and uncompressed or compressed audio signals. The advanced version of HDMI, such as 2.0, can transfer video signals of up to a resolution of 4096x2160.


PS/2 port
(keyboard)
USB ports
Serial port
(dial-up modem)
VGA port
(monitor)
Speakers
Line In
Microphone
PS/2 port
(mouse)
Ethernet port
(network)
LPT1 Printer port
(printer)
Game port
(joystick)